The UG Program at IIT Kanpur

Interim Report

Academic Programme Review Committee

February 2009

Introduction

- 50 years of IIT Kanpur
- India is a global economic power
  - Aspirations of the society have increased
  - Demands on industry have drastically changed
- Address major paradigm shifts
  - Globally competitive
  - Rapidly evolving technological needs
  - Rising expectations of students and parents

Background

- Understand evolving needs of stakeholders
  - **Students**
  - **Teachers**
  - **Industry**
  - **R&D**

- **Students**
  - IITs attract the cream of the country's students
  - Black box approach
    - Motivational deficit
    - Lack of intellectual curiosity
  - Poor communication skills
  - Ignorance of world around

- **Teachers**
  - Highly dedicated and knowledgeable faculty
  - Teaching style
    - Packed content
    - Frenetic pace of instruction
    - Inspirational teaching?
  - Overemphasis on analysis
    - Scope for tapping students' creative potential limited
    - Very little synthesis
    - Inadequate connection to real engineering problems

- Industry/R&D
  - Significant value addition
  - Cutting edge technologies
  - Vision
  - Adapt technology to changing environment
  - Address law of diminishing returns in technology development

Focus

- Carve a niche in Science & Technology
  - Provide avenues to hone skills
    - Managerial
    - Entrepreneurial
    - Social
- Technological solutions to local problems
- Nation building through technology development
  - Sensitise students to local needs
  - Inculcate spirit of team work
  - Cultivate leadership qualities
  - Instill national pride

Product

- Humane, global Indians who are leaders in their field

Skill-Sets

- Basic skills
  - Mathematical skills
  - Analytical skills
  - Experimental skills
  - Hardware friendly
  - Scientific temper
  - Communication skills
  - Cultural and social awareness

- Advanced skills
  - Department-oriented skills
  - Interdisciplinary orientation
  - Synthesis

- Auxiliary skills
  - Management
  - Entrepreneurship

Current Curriculum

- Strengths
  - Science-based education
  - Emphasis on fundamentals
  - Rigorous analysis
  - Humanities and social sciences

- Weaknesses
  - Less emphasis on working with hands and tinkering
  - Too much content
  - Lack of flexibility to select courses vis-a-vis maturity, aptitude and interest
  - Little scope for inter-disciplinary specialization
  - Little stress on communication skills in evaluation

Proposed Curriculum

- Truly credit-based curriculum
- Develop aptitude for experiment and exploration
  - Project-based laboratories
  - Minimum laboratory credits
- Conceptually challenging and thought-provoking curriculum
- Emphasis on inter-disciplinary education

- Cater to good, motivated student
  - UG research
  - Earning credits with universities of repute
  - Flexibility to take more courses through overload
  - Project-oriented internships in industry/R&D laboratories for credit

Credits

- Present
  - Academic Load (AL) = 3L + T + 1.5P + 0DH
  - $5 \leq AL \leq 15$
  - Typical AL = 10 – 11
  -

AL	Weightage
5 – 6	2
7 – 8	3
9 – 12	4
13 – 15	5

- Proposed
  - Students expected to spend average 50 – 55 h/week, including exam/quiz preparation
  - Credits should reflect contact and self-study hours spent by students
  - Drop discussion hour
  - C = L + T + P + SS
    - SS = 2L + T + A, where $0 \leq A \leq 2$
  - 400 – 420 credits for graduation

Graduation

- Present
  - Passing grade = D (4)
  - Graduating CPI = 5.0

- Proposed
  - Clear approximately 400 – 420 credits
  - Passing grade = D (4)
  - Graduating CPI = 4.0
    - Warning: CPI < 4.5
    - Termination: CPI < 4.0
  - De-motivate deficient students
    - Minimum attendance requirement

Degrees

- Present
  - B.Tech. (4 yrs)
  - Integrated MSc (5 yrs)

- Proposed
  - All first degrees of 4 yrs duration
    - B.Tech. (Engg.) and B.S. (Sciences/Economics)
  - Option for Minors/IDM in a second branch (incl. HSS)
  - One more year for the second degree
  - Option for a second degree exercised at the end of III year
    - M.Tech. (Engg.), MSc (Sciences/Economics), MBA
    - Second B.Tech. degree
  - Switching between Science/Economics and Engg. degrees

- B.Tech. in Engineering Sciences
  - Initially this branch will be offered as a “branch change” option.
  - It will have a compulsory core like other students.
  - Stress will be on engineering sciences.
  - Multiple minors or interest-based course stream in departmental compulsory slots

Academic Programme Structure

	HSS	Sc	Esc	TA	Deptt	Open/Core Electives
Original 1963	18.7	22.6	13.2	11.3	34	
I UGRC 1970	16	20	10	10	32	12
II UGRC 1981	10.9	12.9	15.2	6.5	41.3	12.9
III UGRC 1992	10	15	15	5	42.5	12.5
IV UGRC 2001	9.6	19.2	9.6	2.4	50.0	2.4
V ARC 2008 Proposed	**12-15 Includes Mgt.+ Comm Skills+ Foreign Lang.**	**20 *Compulsory Electives***	**10**	**2-5 Manufact. + Engg. Drawing**	**35-40 Comulsory : 25-30 Electives: 10**	**15**

Core

- Broad-based courses
- Flexible core programme:
  - Compulsory and elective components
  - Fixed slots for compulsory courses
  - Departments decide slots of some courses
  - Students free to choose slots for electives
  - Specified core credits for first 2 years
- Multi-departmental participation mandatory
- PE offered in first 2 semesters
  - Faculty instructors/tutors; coaches assist tutors

- Sciences
  - Divide into compulsory and elective slots
  - Compulsory courses cater to all departments
  - Electives contain department-specific core courses
  - Compulsory curriculum
    - Basic
      - 2 Maths, 2 Physics, 1 Chemistry, Physics and Chemistry laboratories
    - Exposure to emerging sciences
      - Introduction to Biology

- Esc
  - Retain Electronics and Programming with modifications
  - Programming
    - Incorporate more “scientific computing”
    - Need to deal with students with below threshold exposure
  - Electronics
    - Reduce course content
    - Simplify and pitch it to an average student
    - Laboratory should be fun
      - Introduction of projects (fabricate an amplifier)
    - Move to II year because of math pre-requisite

- ESO
  - Should not be department-centric
  - Comfort level of students from participating departments

- TA
  - Drawing
    - Do away with drafting
    - Sketching and computer-based
  - Manufacturing
    - Addition of old TA203 + TA204
    - Very heavy: (2Lecture + 2Practical)/week + 2 projects
    - Split course
      - Two independent courses: ME and MME (1L + 1P)/week
      - Theory taught in I year; Both practical together in II year

- HSS
  - Regular courses as before
  - Communication skills
  - General management
  - Foreign languages
  - Courses on Indian, world history

- Modular courses
  - Half-semester courses with half credits
  - Flexibility for department-specific core
    - PDE, Complex analysis, Waves, Lasers
  - Introduce new, specialised short-courses
    - Fracture mechanics, Smart materials, NDT,Chip design
  - Eliminate “stretched” courses
  - One mid-semester and end-semester examinations

Implementation

- Students
  - Lack of vibrant academic atmosphere
  - Inculcate a sense of balance among students
    - Minimum attendance requirements
  - Number, scale, and funding of festivals should not relegate “academics”
  - Increase interest of students in experiments
  - Curb rampant copying in laboratory reports and home work

- Faculty
  - Balance between (i) teaching, (ii) corporate activities, and (iii) research
  - Course syllabus and instruction followed in letter and spirit
    - Maintain course files
    - Form course monitoring groups
- Fixed time table
- Examination and evaluation
  - 2 mid-semester vs 1mid-semester examinations

Thank You!